

Repair Manuals

Timing chain: removal/installation, (03/2012 -)

Warnings and recommendations

Before working on the fuel system, clean the area thoroughly

Always use new gaskets

Removal

Disconnect the battery earth cable

Remove the windscreen wiper arms

Remove the dust and pollen filter housing

Remove the cowl top panel

Remove the strut brace

Note: Right-hand side

Remove the engine upper covers

Remove the bulkhead cover

Disconnect the engine wiring from the top of the engine

Remove the ignition coils

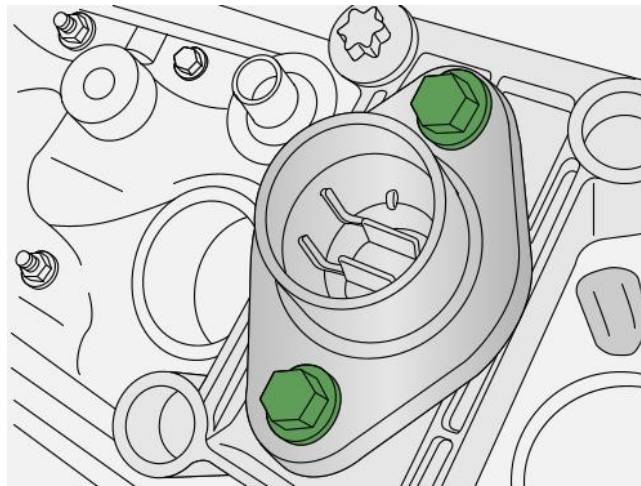
Reduce the pressure in the fuel system

Remove the injector lines

Note: Plug the ends of the fuel lines

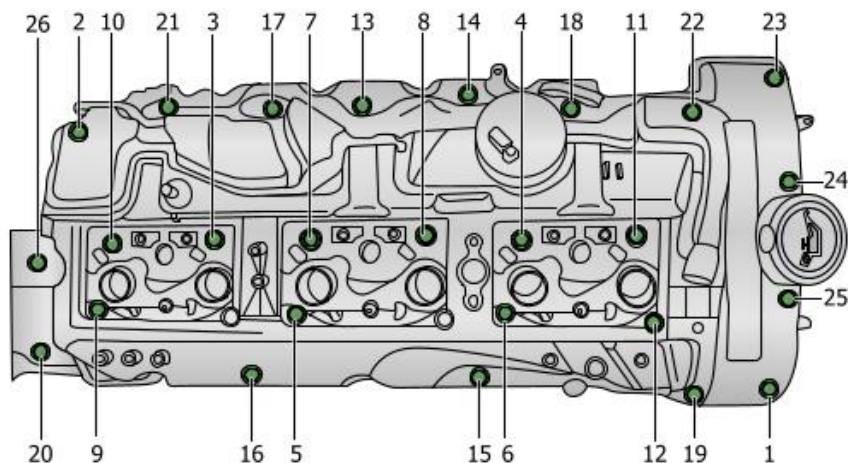
Remove the air filter outlet hose

Remove the flange



Disconnect the vacuum hose from the valve cover

Loosen the bolts 1/2 a turn at a time, to prevent damage. From outside to inside ((26 -> 1)).



Remove the valve cover

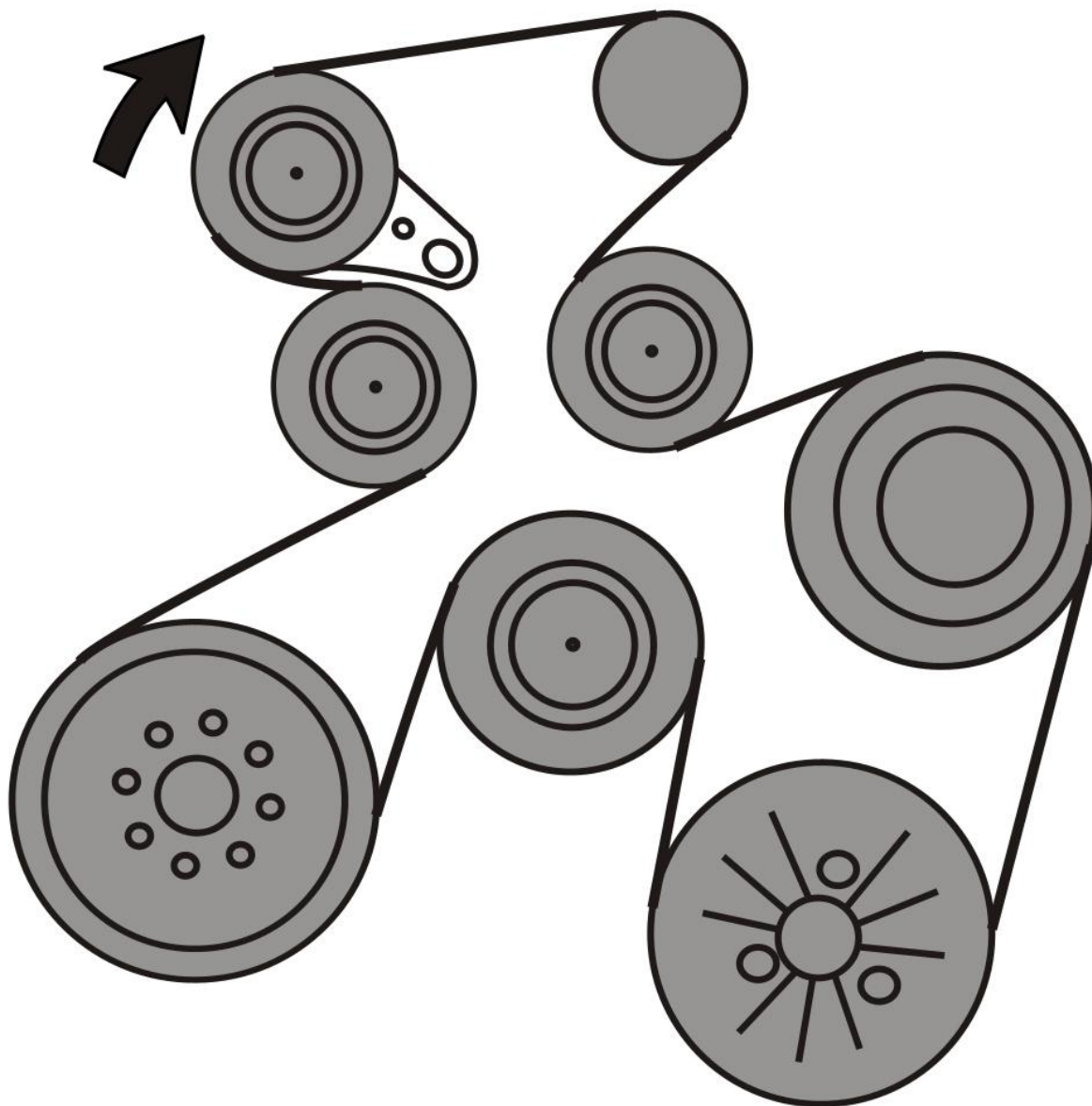
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Remove the spark plugs (12 1 220)

Remove the engine lower cover

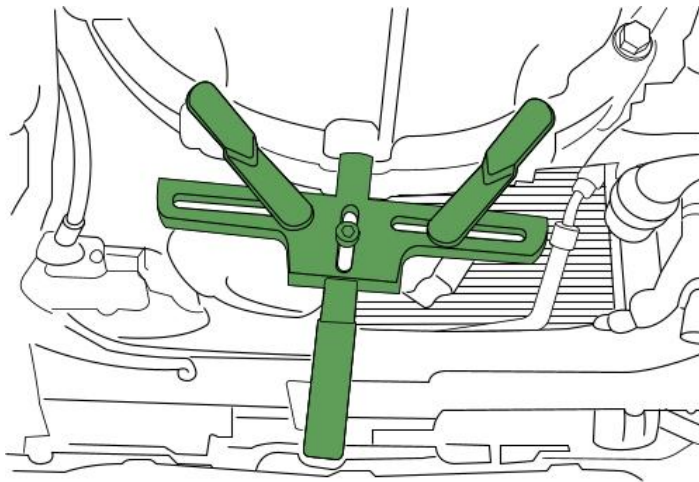
Remove the ancillary drive belt

Drive belt layout (With air conditioning)



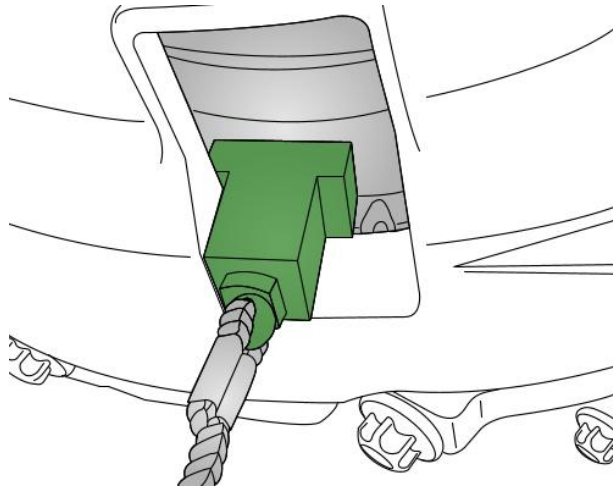
Vehicles with manual transmission:

Fit the crankshaft locking tool (11 8 180)

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Vehicles with automatic transmission:

Fit the crankshaft locking tool (11 8 660)

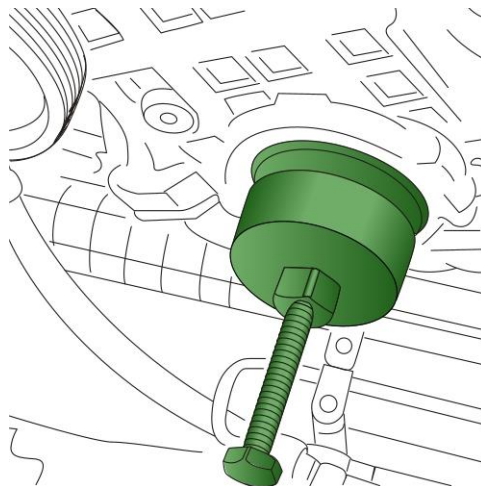


All vehicles:

Remove the crankshaft pulley

Fit the special tools ((80 Nm)) (11 0 371, 11 0 372)

Remove and discard the crankshaft oil seal



Remove the tensioner locking pin (11 0 390)

Remove the ancillary drive belt tensioner

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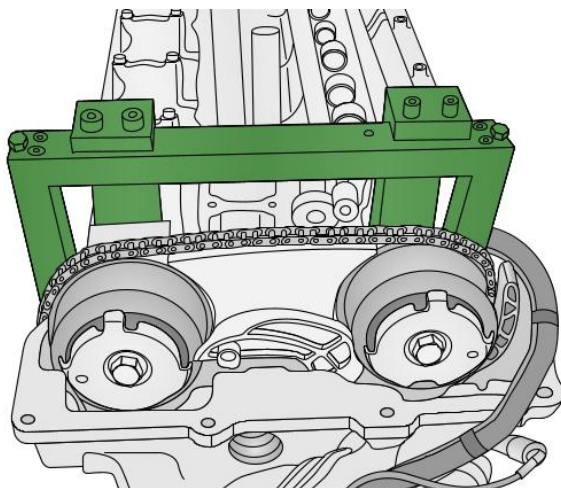
Remove the special tool (11 8 180, 11 8 660)

Check the valve timing

Proceed as follows:

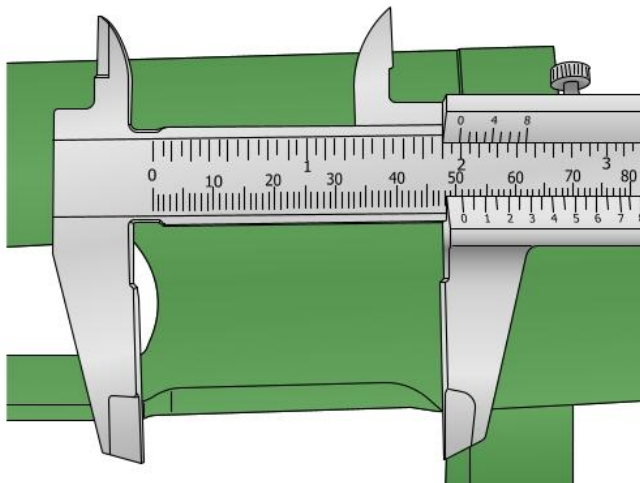
Turn the camshafts to check that the camshaft adjustment units can be locked

Assemble the special tool as shown (11 4 280)

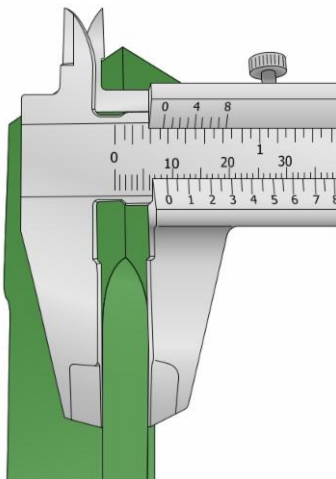


Inlet camshaft:

Modify the special tool chamfer as shown ((50 mm))

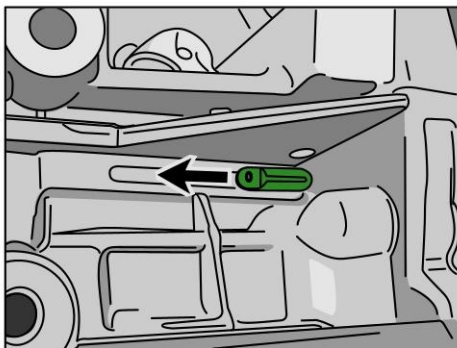


((8.5 mm))



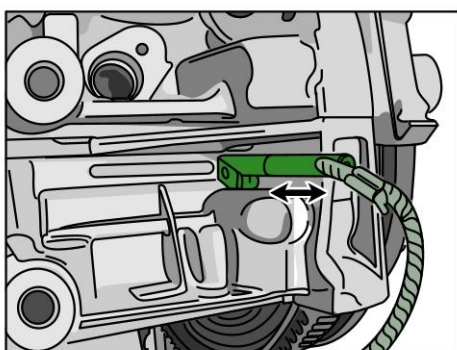
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Remove the access plug



Turn the crankshaft to TDC for cylinder 1

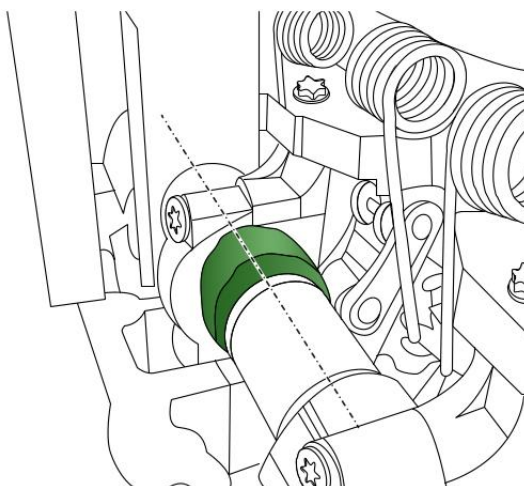
Fit the crankshaft TDC locking pin (11 0 300)



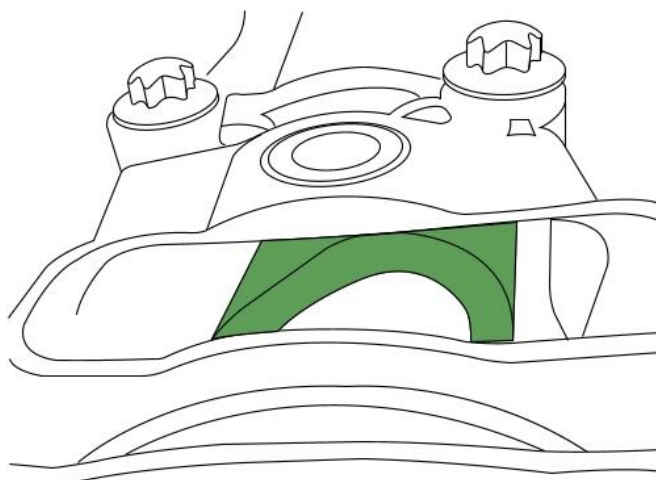
Ensure the crankshaft is locked

Check the position of the camshafts

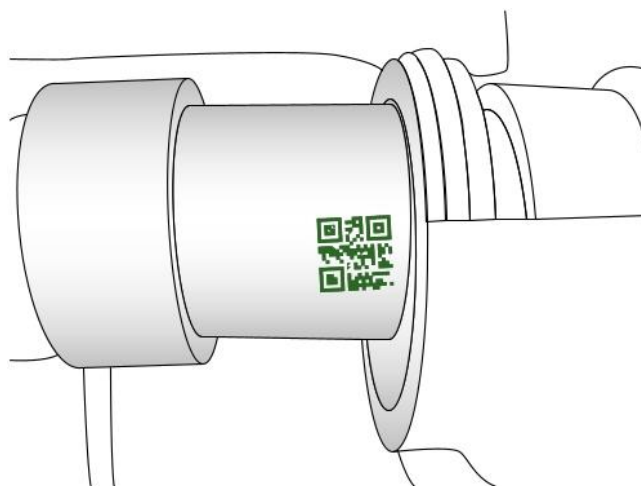
Inlet camshaft:



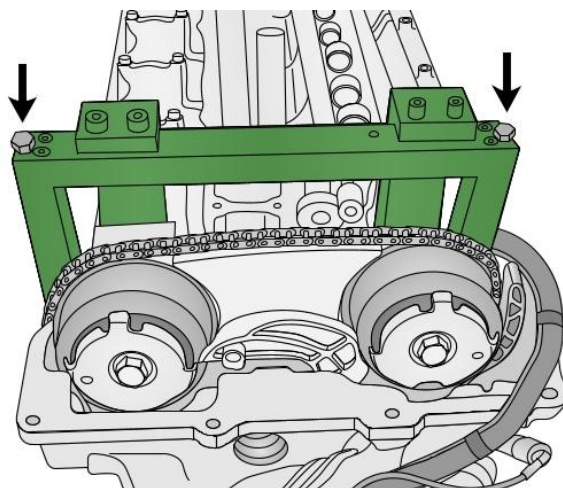
Exhaust camshaft:

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Note: Ensure that the marks face upwards

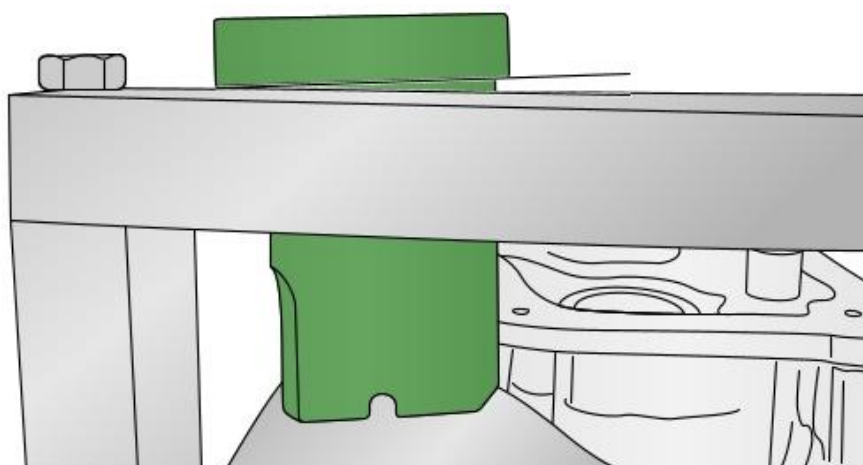


Secure the special tool with two bolts (11 4 285, 11 4 282, 11 4 281)

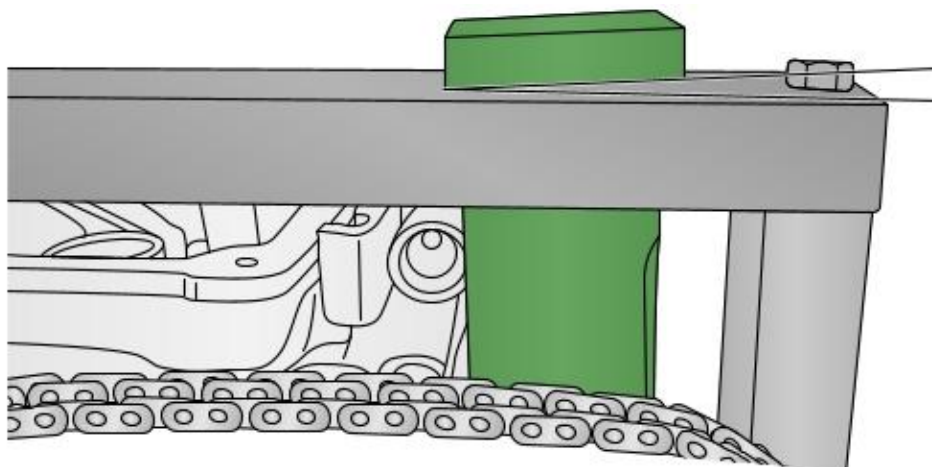


Check for correct fitting

Exhaust side:

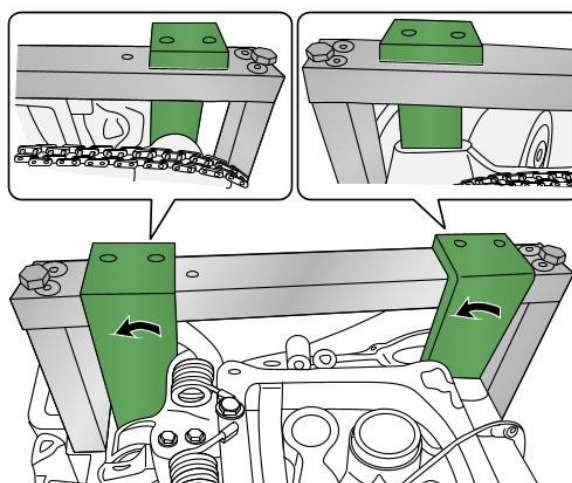
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Inlet side:



Position the special tools as shown

Note: Ensure correct fitting

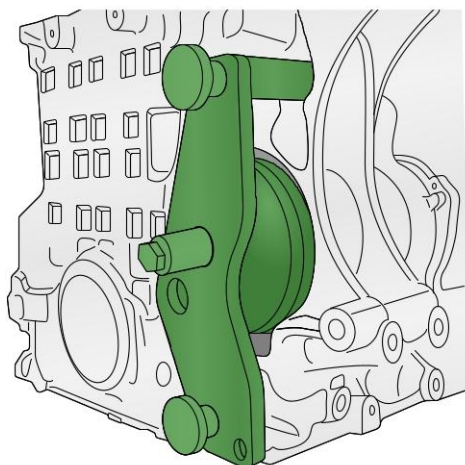


Vacuum pump:

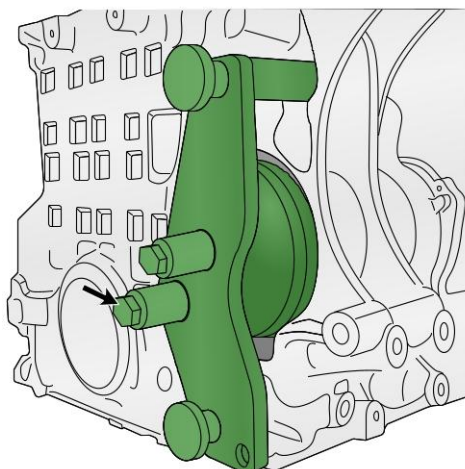
Assemble the special tool as shown (11 8 530)

With washer: ((6 mm))

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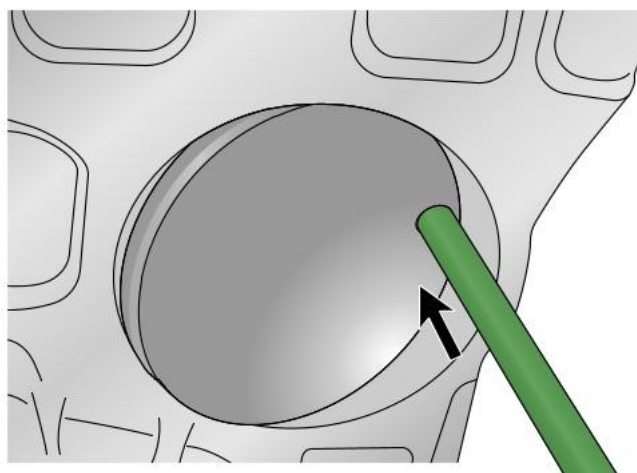
Tighten the special tool
Remove the sealing cap



Oil pump chain tensioner:

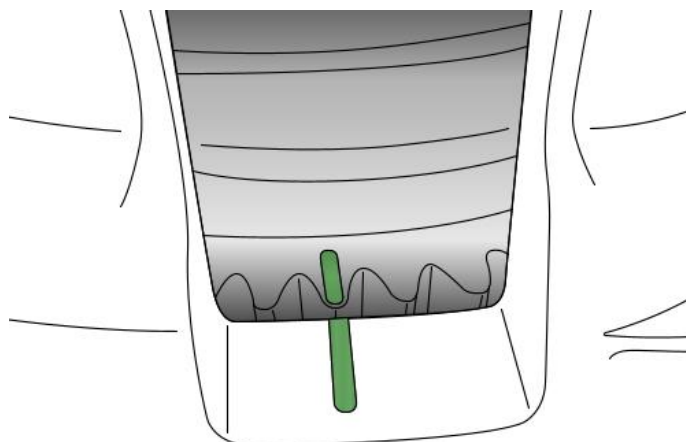
Remove the sealing cap

Note: Use a punch



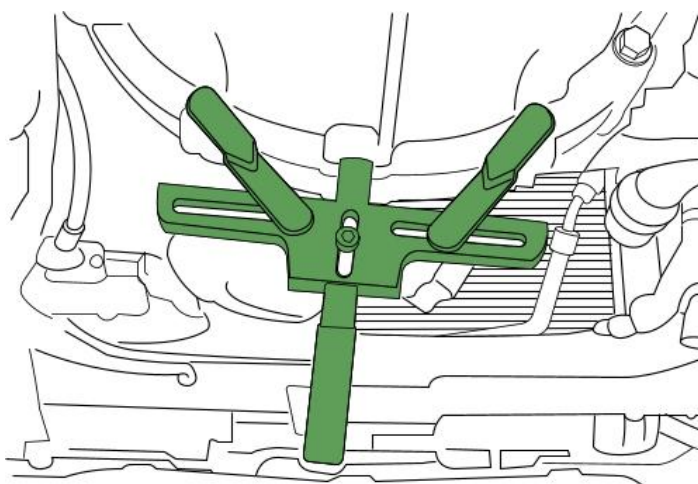
Remove the crankshaft TDC locking pin (11 0 300)

Note: Mark the position

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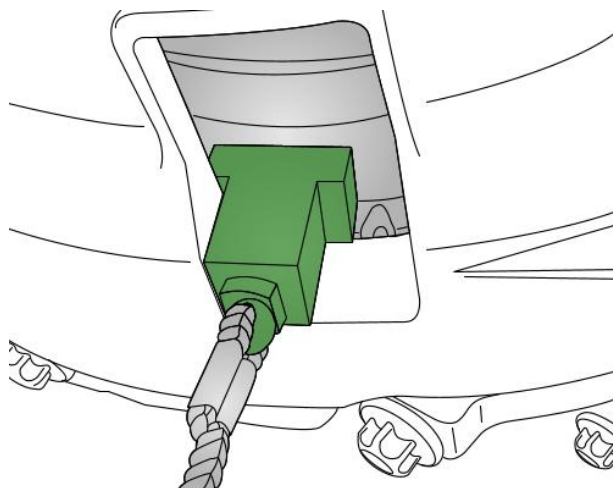
Vehicles with manual transmission:

Fit the crankshaft locking tool (11 8 180)



Vehicles with automatic transmission:

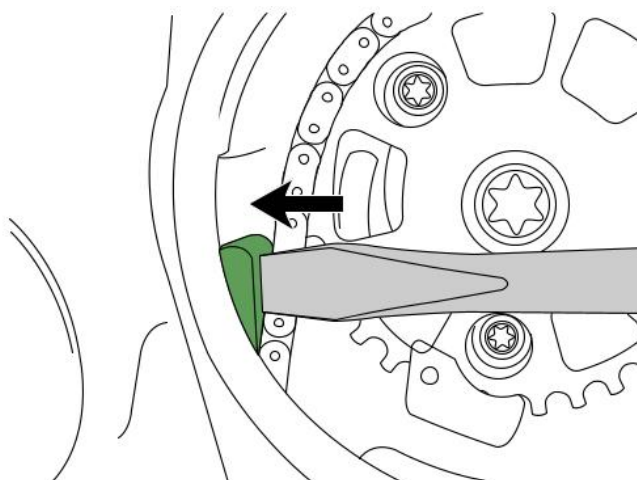
Fit the crankshaft locking tool (11 8 660)



All vehicles:

Apply pressure to the chain to release the tensioner piston (11 4 120)

Repeat the procedure several times if necessary

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Lock the tensioner; use a locking pin

Remove the crankshaft gearwheel centre bolt

Breakaway torque: ($> 600 \text{ Nm}$)

Right-hand side:

Remove the air hose between the turbocharger and the intercooler

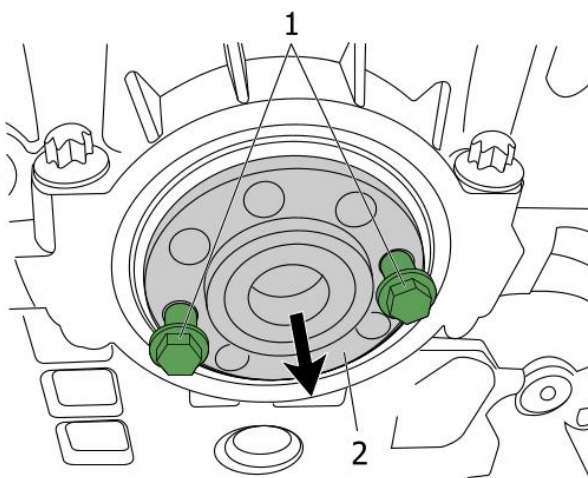
Remove the coolant reservoir, but do not disconnect

Remove the turbocharger air inlet pipe

Remove the chain tensioner

Refit the crankshaft pulley bolts ((1))

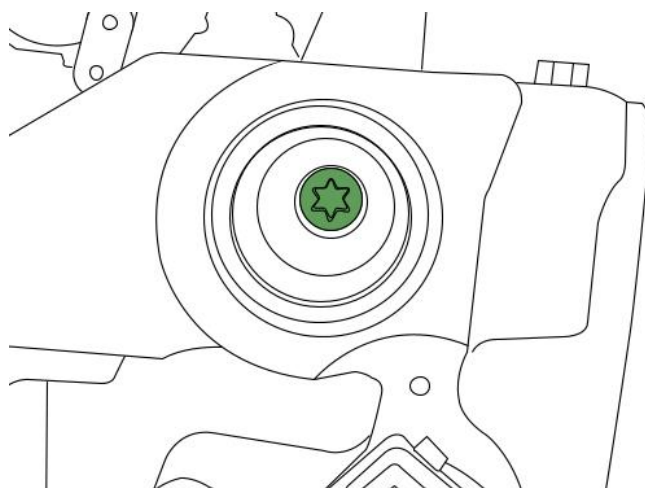
Remove the crankshaft hub ((2))



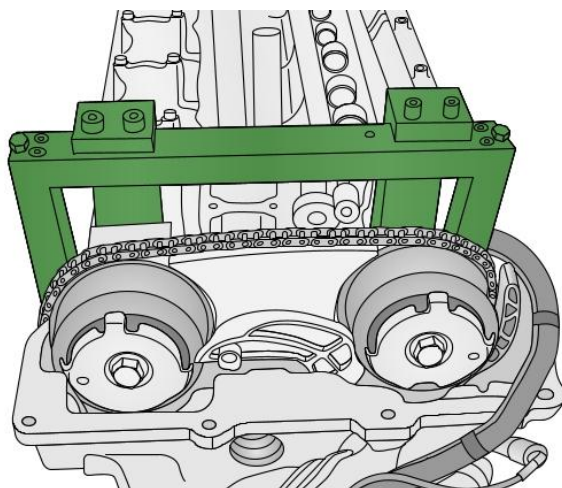
Remove the access plugs

Remove the bearing pin from the cylinder head

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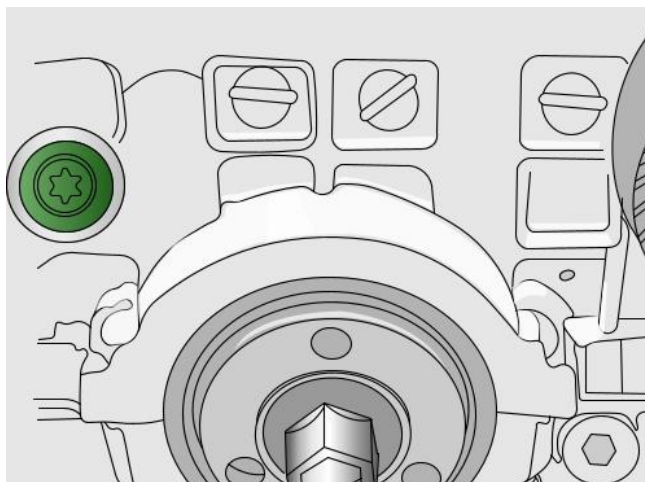
Fit the camshaft locking tool (11 4 280)



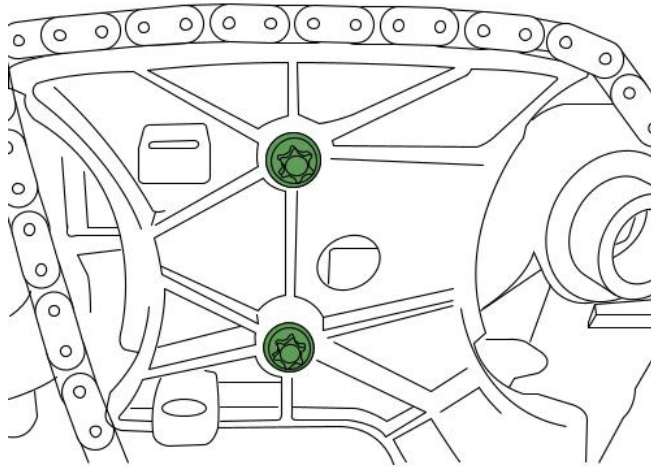
Loosen the camshaft gearwheel bolts

Remove the camshaft gearwheels

Remove the bearing pins for the chain guide rail and the tensioner rail



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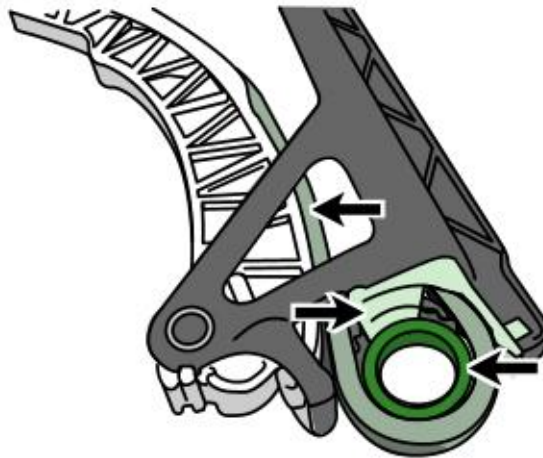
Remove the chain guide with the timing chain, guide rail and tensioner rail

Installation

Refit the chain guide with the timing chain, guide rail and tensioner rail

Check for correct fitting

Tension the chain



Refit the bearing pins for the chain guide rail and the tensioner rail

Refit the timing chain guide bolts to the cylinder head

Refit the crankshaft hub

Tighten the crankshaft gearwheel bolt (00 9 140)

Check the camshaft gearwheel markings

Exhaust camshaft gearwheel: ((AUS/EX))

Inlet camshaft gearwheel: ((EIN/IN))

Note: Ensure correct fitting

Refit the camshaft gearwheels

Note: Use new bolts

Tighten the camshaft gearwheel bolts lightly until there is no play

Loosen the gearwheel bolts ((90°))

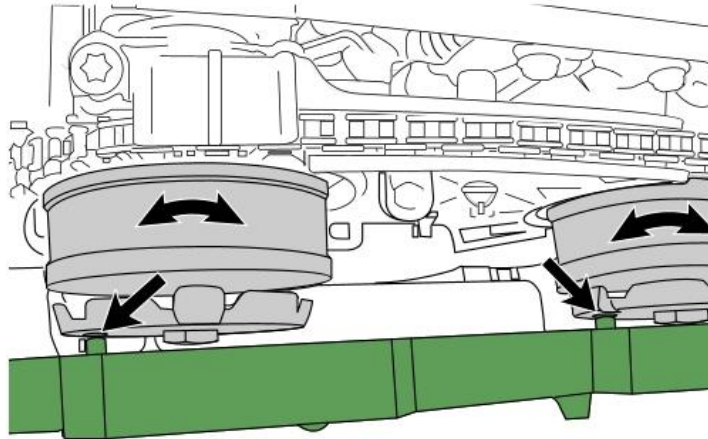
At the same time press and hold the tensioner rail

Check that the special tool is correctly fitted (11 4 280)

Fit the camshaft alignment tool (11 4 290)

Rotate the camshaft sensor wheels until the retaining pins on the special tool fit correctly

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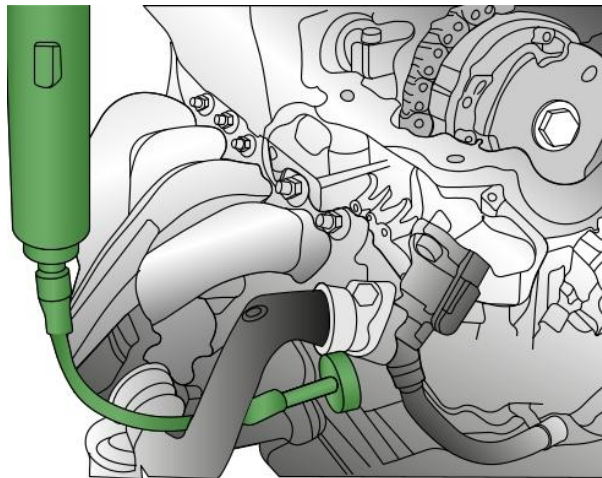


Secure the special tool with two bolts (11 4 290)

Fit the dummy chain tensioner (11 9 340)

Pre-tension the timing chain ((0.6 Nm))

Use the special tool (00 9 250)



Tighten the camshaft gearwheel bolts

Remove the special tools (00 9 250, 11 4 290, 11 9 340, 11 4 280)

Compress the chain tensioner piston twice to drain the oil

Refit the chain tensioner

Note: Renew the sealing ring

Turn the engine 2 rotations clockwise

Check the timing

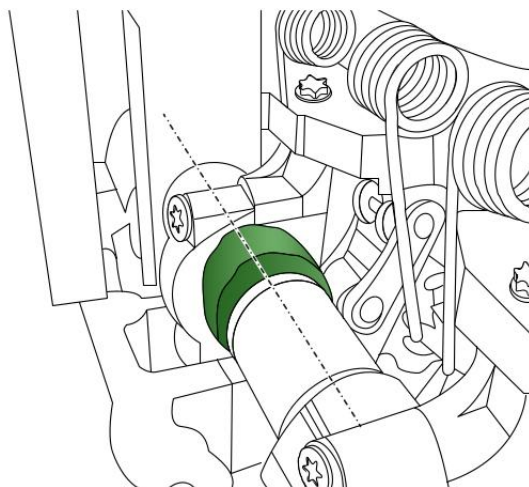
If the special tools do not fit, adjust the valve timing (11 4 280)

Proceed as follows:

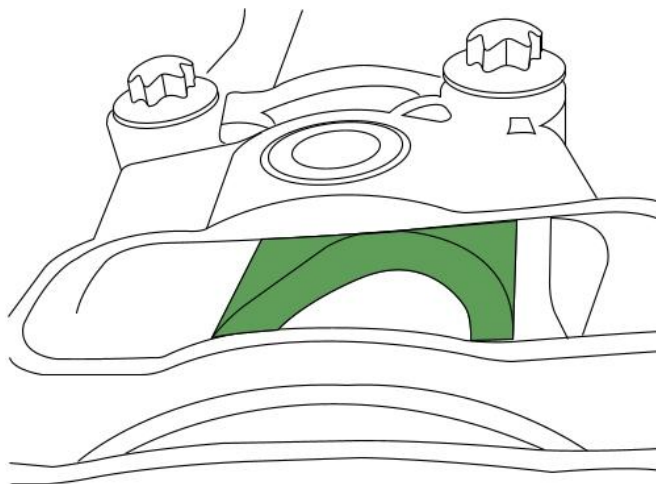
Check the position of the camshafts

Inlet camshaft:

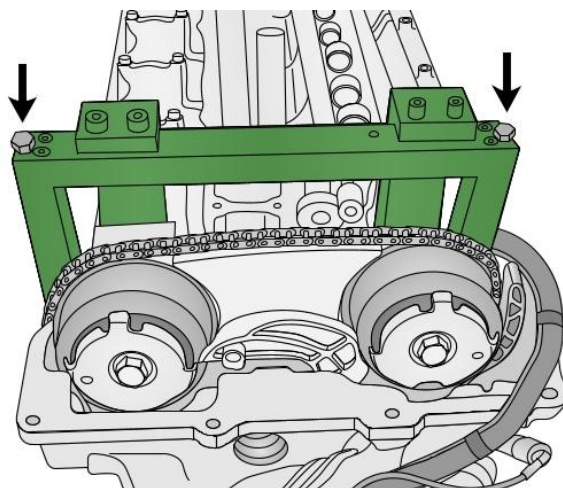
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Exhaust camshaft:



Secure the special tool with two bolts (11 4 285, 11 4 282, 11 4 281)



Check that the camshafts are locked

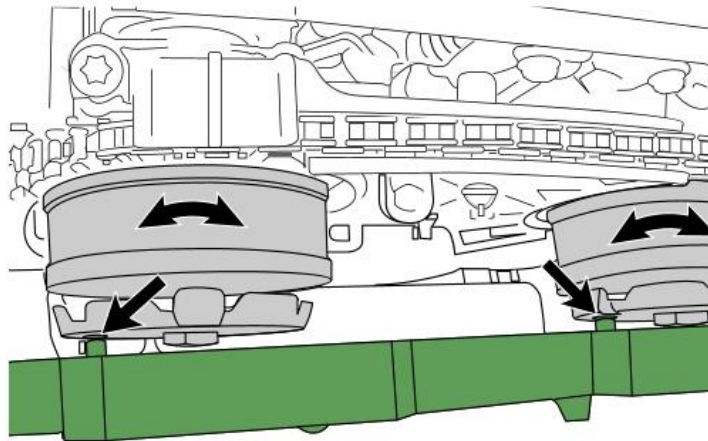
Loosen the camshaft gearwheel bolts

Remove the chain tensioner

Fit the camshaft alignment tool (11 4 290)

Rotate the camshaft sensor wheels until the retaining pins on the special tool fit correctly

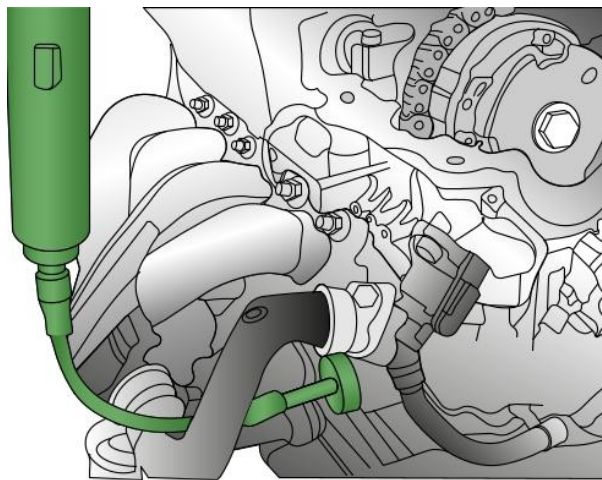
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Secure the special tool with two bolts (11 4 290)

Pre-tension the timing chain ((0.6 Nm))

Use the special tools (00 9 250, 11 9 340)



Tighten the camshaft gearwheel bolts

Remove the special tools (00 9 250, 11 4 290, 11 9 340, 11 4 280)

Refit the chain tensioner

Turn the engine 2 rotations by hand

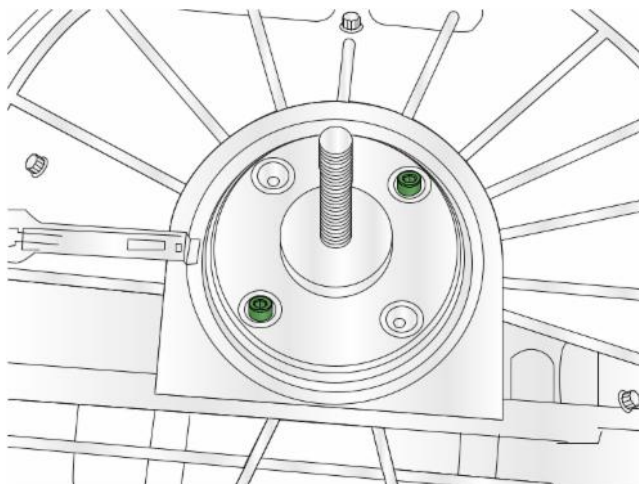
Check the timing again

The timing is set when the special tools fit correctly

Fit the crankshaft seal over the special tool (11 9 235)

Note: Clean off the sealing surfaces

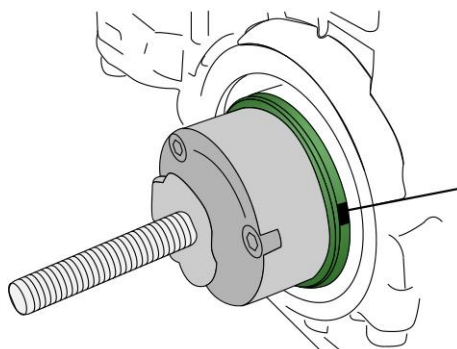
Fit the special tool. Secure the special tool with two bolts.

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Check the seal

Align the marks

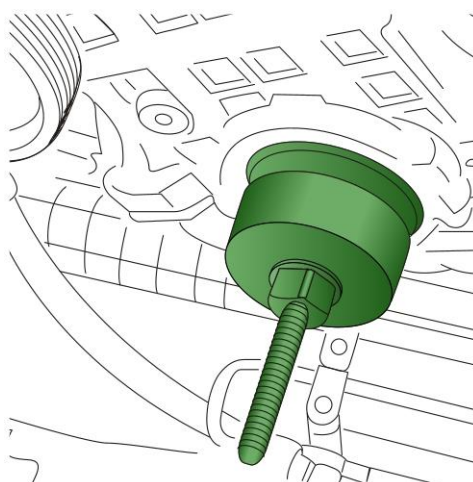
Note: Ensure correct fitting



Fit the special tool (11 9 231)

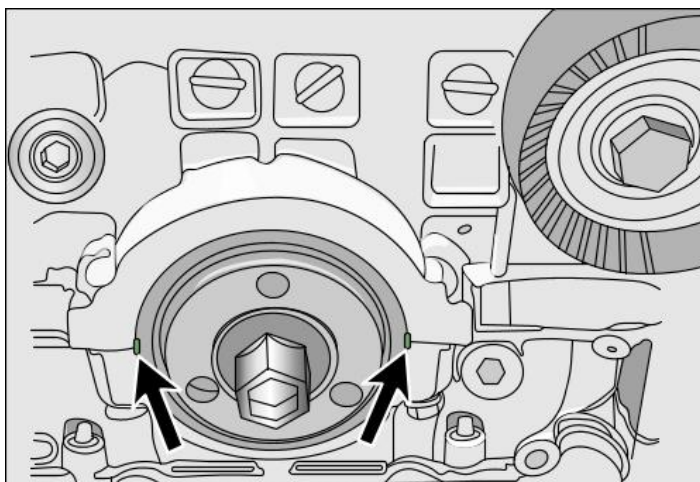
Tighten the special tool

Note: Until the crankshaft seal is level with the timing cover



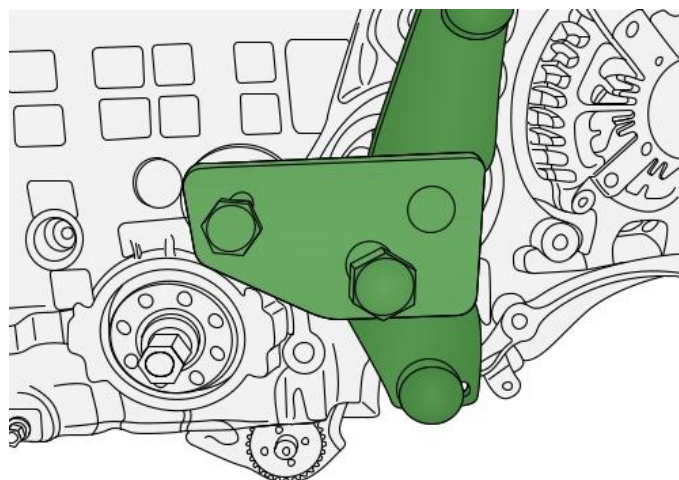
Remove the special tools (11 9 235, 11 9 231)

Apply sealant as indicated. Loctite: ((171000)).

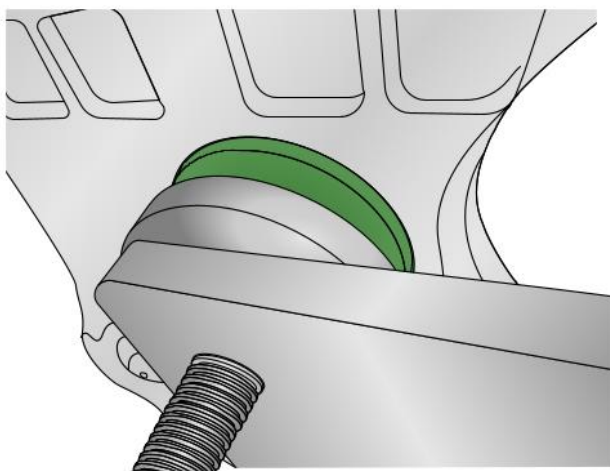
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Oil pump chain tensioner:

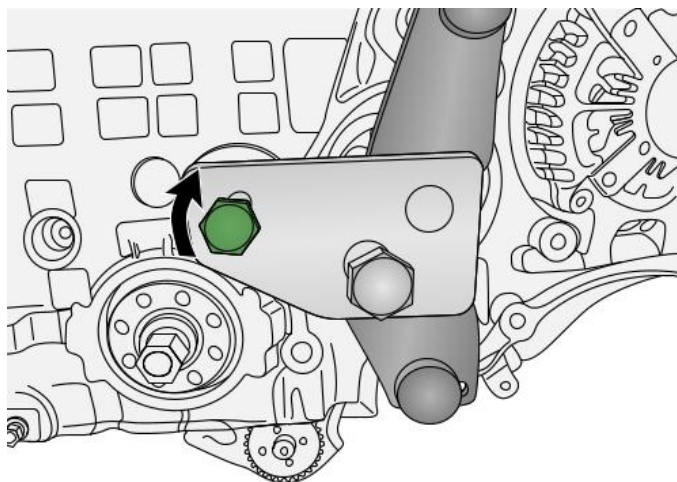
Assemble the special tool as shown (2 318 118, 2 318 119, 11 8 531, 11 8 532, 11 8 535)



Fit the sealing cap

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Tighten the bolt ((20 Nm))

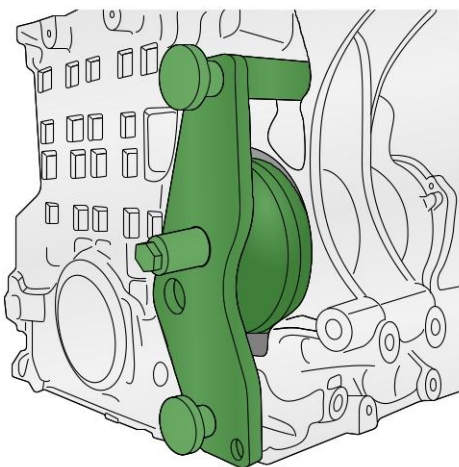


Remove the special tools (2 318 118, 2 318 119, 11 8 531, 11 8 532, 11 8 535)

Vacuum pump:

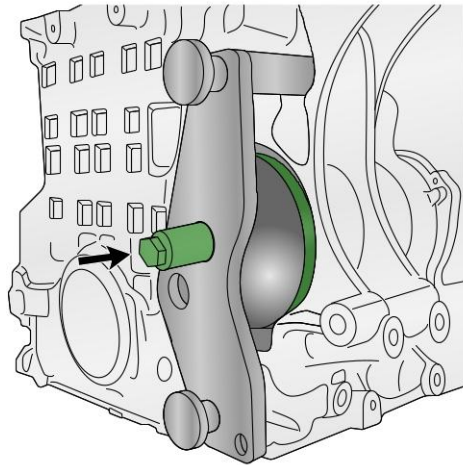
Fit the sealing cap

Fit the special tools (11 8 531, 11 8 532, 11 8 533)



Tighten the bolt

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Remove the special tools (11 8 531, 11 8 532, 11 8 533)

Further refitting is in the reverse order of removal

Reconnect the battery earth cable

Torque settings

Valve cover

Valve cover 8.5 (Nm)

Spark plugs

Spark plugs (M12) 23 ±3 (Nm)

Spark plugs (M14) 30 ± 3 (Nm)

Crankshaft pulley

Crankshaft pulley (Outer bolts) 35 (Nm)

Crankshaft gearwheel

Crankshaft gearwheel (Centre bolt)

Renew the bolt (Threads lightly oiled)

Stage 1 100 (Nm)

Stage 2 270 (°)

Timing chain tensioner

Timing chain tensioner (Upper)

Renew the seal

Stage 1 55 (Nm)

Camshaft gearwheel

Camshaft gearwheel

Renew the bolts

Stage 1 20 (Nm)

Stage 2 180 (°)

Timing chain guide

Timing chain guide (Upper) 14 (Nm)

Timing chain guide (Lower) 20 (Nm)

Strut brace:

Use new bolts

M10:

Stage 1: (40 Nm)

Stage 2: (60°)

M12:

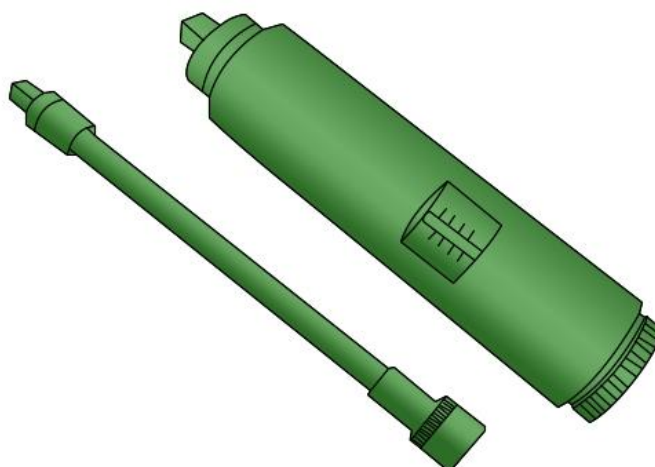
Stage 1: (100 Nm)

Stage 2: (100°)

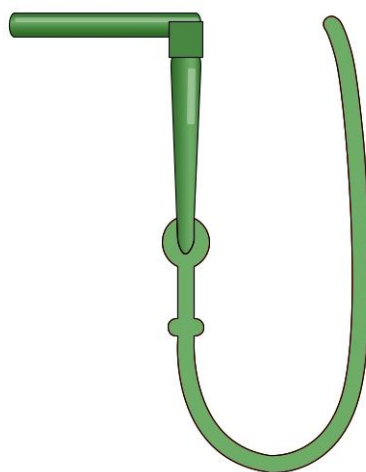
Special tools

Adjustment tool: (00 9 250)

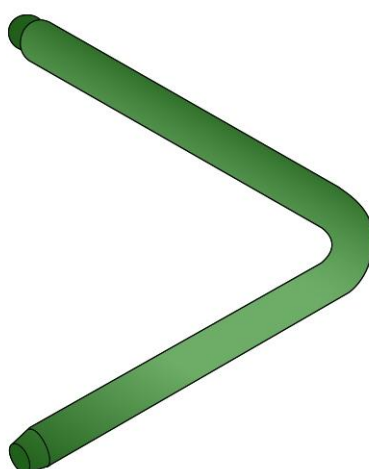
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Crankshaft locking tool: (11 0 300)

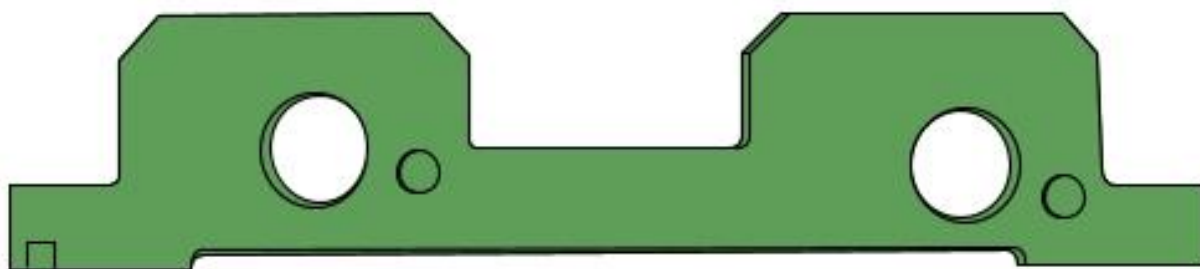


Tensioner locking pin: (11 0 390)

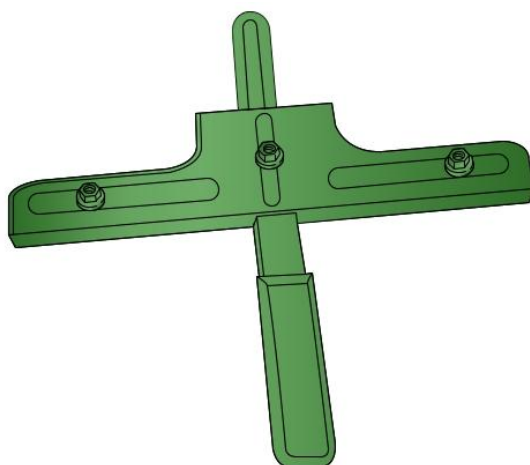


Camshaft alignment tool: (11 4 290)

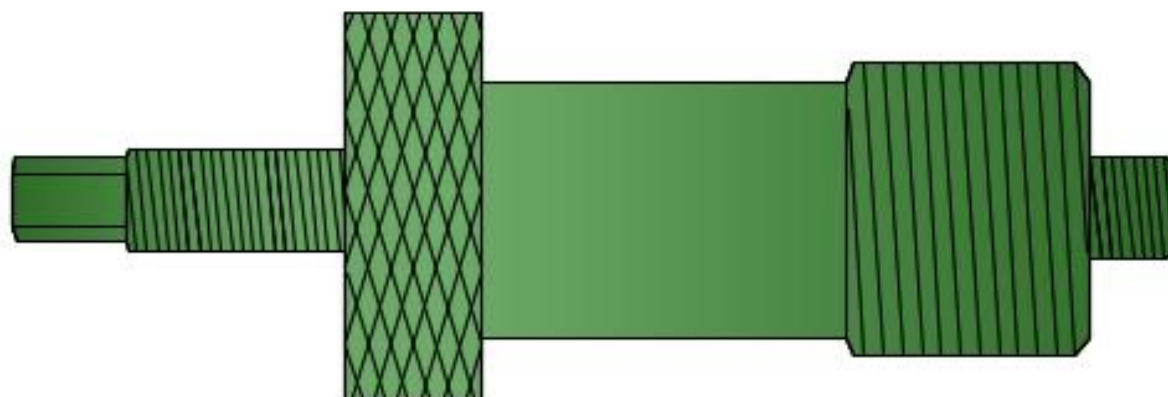
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Crankshaft locking tool: (11 8 180)

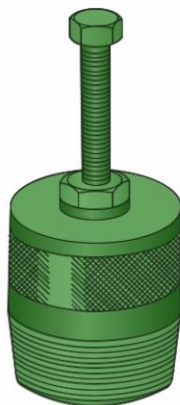


Adaptor: (11 9 340)

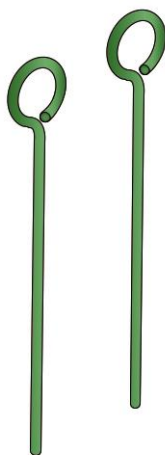


Extractor: (11 0 371, 11 0 372)

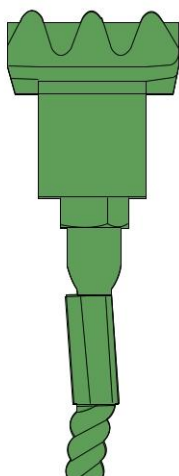
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Locking pin: (11 4 120)

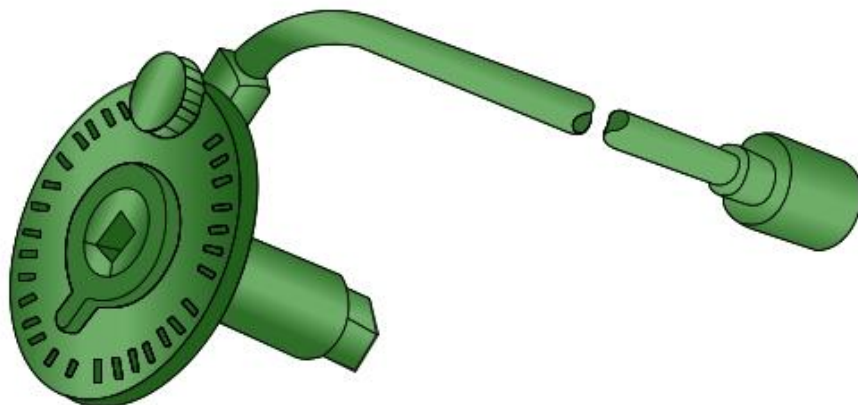


Flywheel locking tool: (11 8 660)

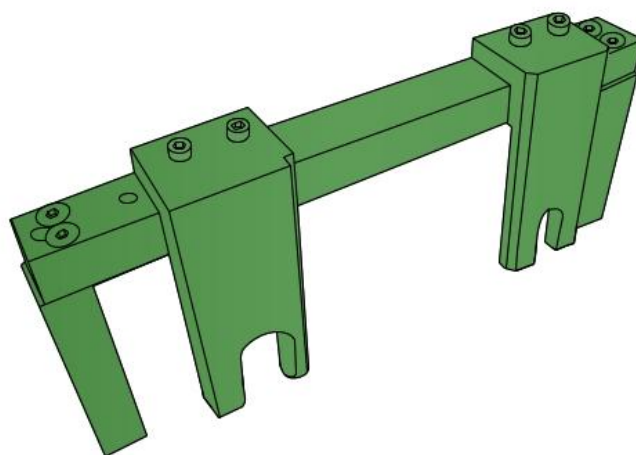


Torque wrench: (00 9 140)

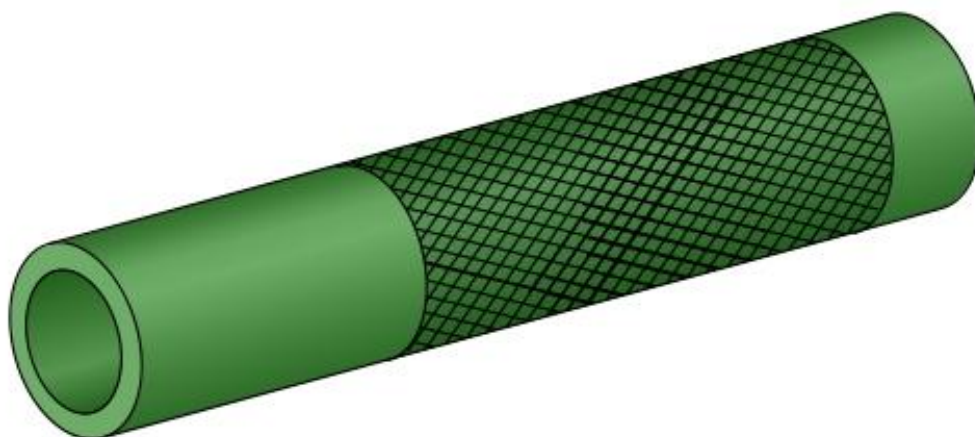
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Camshaft locking tool: (11 4 280)

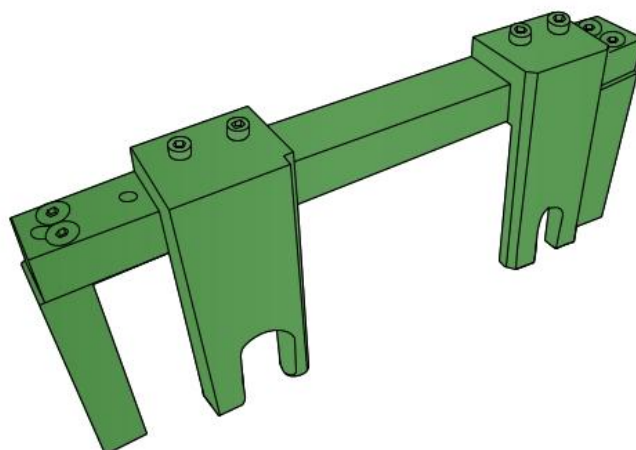


Spark plug removal tool: (12 1 220)

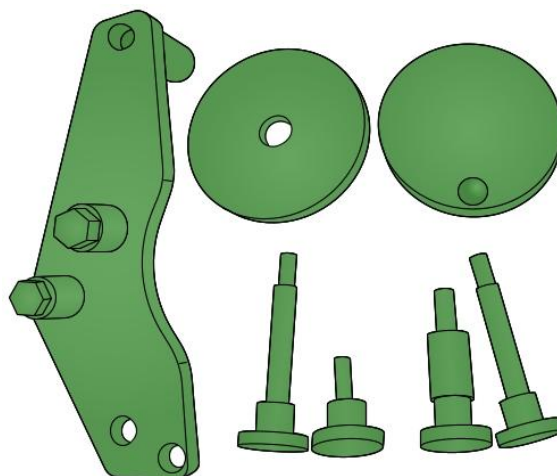


Camshaft timing tool: (11 4 285, 11 4 282, 11 4 281)

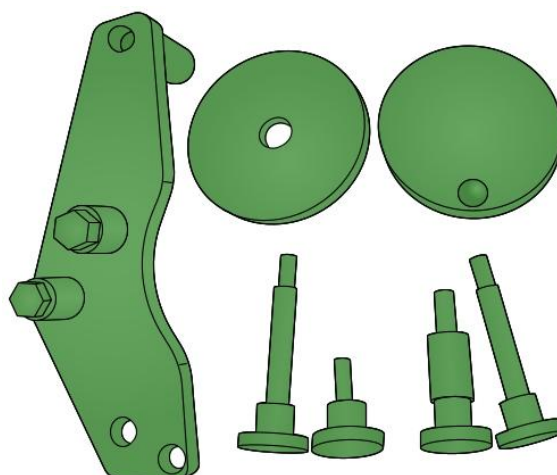
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Special tools: (11 8 530)

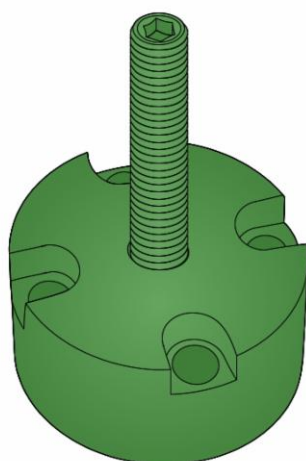


Special tools: (11 8 531, 11 8 532, 11 8 533)

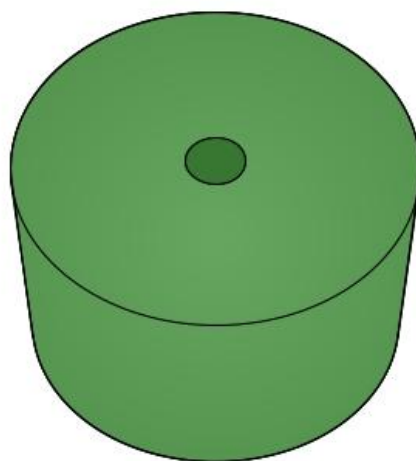


Oil seal fitting tool: (11 9 235)

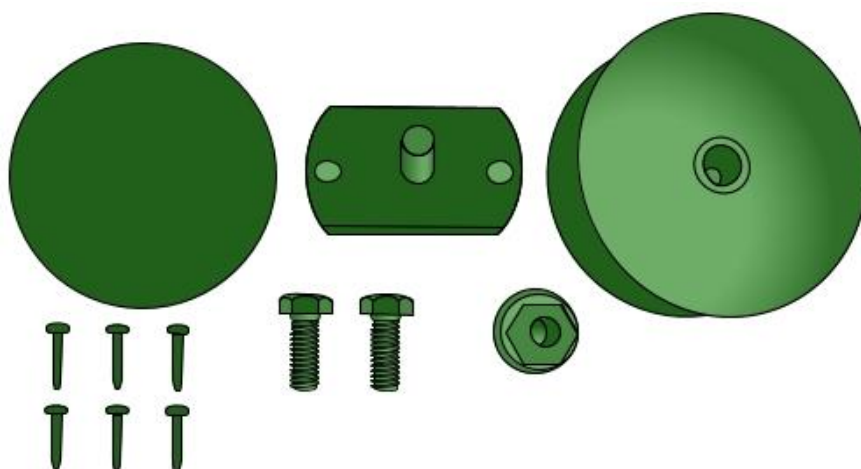
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Crankshaft front radial seal fitting tool: (11 9 231)

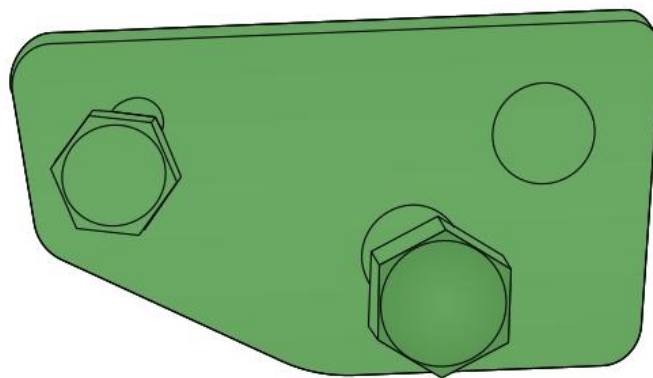


Bolts: (11 9 234)



Special tools: (2 318 118, 2 318 119)

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Special tools: (11 8 531, 11 8 532, 11 8 535)

